

Chapter 52. Airway Secretions and Suctioning

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Airway Secretions and Suctioning: Introduction

Retention of airway secretions in patients on invasive mechanical ventilation constitutes a common problem associated with several complications. Therefore, secretion management represents a great challenge for respiratory therapists, nurses, and physicians, particularly in the patient with underlying airways disorders. This chapter discusses management of airway secretions in mechanically ventilated patients, focusing on physiology of mucus production, clearance, and the most common treatments, both pharmacologic and nonpharmacologic, to enhance removal of retained secretions. Although adequate humidification of the inspired gas plays a primary role on mucus clearance, the reader is referred to [Chapter 51](#) for a comprehensive discussion of this topic.

Physiology of Airway Secretion and Clearance

The lungs are exposed daily to 10,000 to 12,000 L of inhaled air, which potentially carries pathogens and noxious particles. Nevertheless, airways are efficiently protected by the airway lining fluid, which constitutes a physical barrier and a medium with antimicrobial and immunomodulatory properties. The airway lining fluid is a biphasic layer formed by a gel-phase (mucus), mainly consisting of water (97%), with proteins, lipids, electrolytes, and cellular debris making up the remaining 3% of its weight. By contrast, the inner layer is a low-viscosity sol-phase that mainly provides lubrication for continuous ciliary beating.

The main components of mucus are mucins. These are large (up to 3×10^6 Daltons per monomer) heavily glycosylated proteins that provide a tangled network necessary to entrap any particles inhaled during ventilation. Two different types of mucins have been identified: (a) cell-tethered and (b) secreted mucins. Cell-tethered mucins present several functions in cellular adhesion, pathogen binding, and cellular transduction. We focus on secreted mucins, because detailed description of membrane-bound mucins goes beyond the scope of this chapter.

Four genes coding for airway-secreted mucins have been identified and [map](#) to chromosome 11p15.5¹ and 12q12.² As depicted in [Table 52-1](#), the genes can be found in different organs; [MUC5AC](#) and MUC5B are the ones most commonly expressed in human airways.^{3,4} Mucin genes comprise a single large central domain (approximately 10 kb) that encodes for serine-rich, threonine-rich, and proline-rich regions, which are the sites of *O*-linked glycosylation of mucins. Importantly, linked glycans present an extraordinary diversity⁵ among species and even within species; this feature provides the highly efficient binding capability between mucus and almost any particle and/or pathogen deposited on the airways. The 5' and 3' genomic regions encode for the von Willebrand factor-like proteic regions, rich in cysteine,⁶ which allow the disulfide bonds between mucin monomers to form the ultimate polymeric structure.

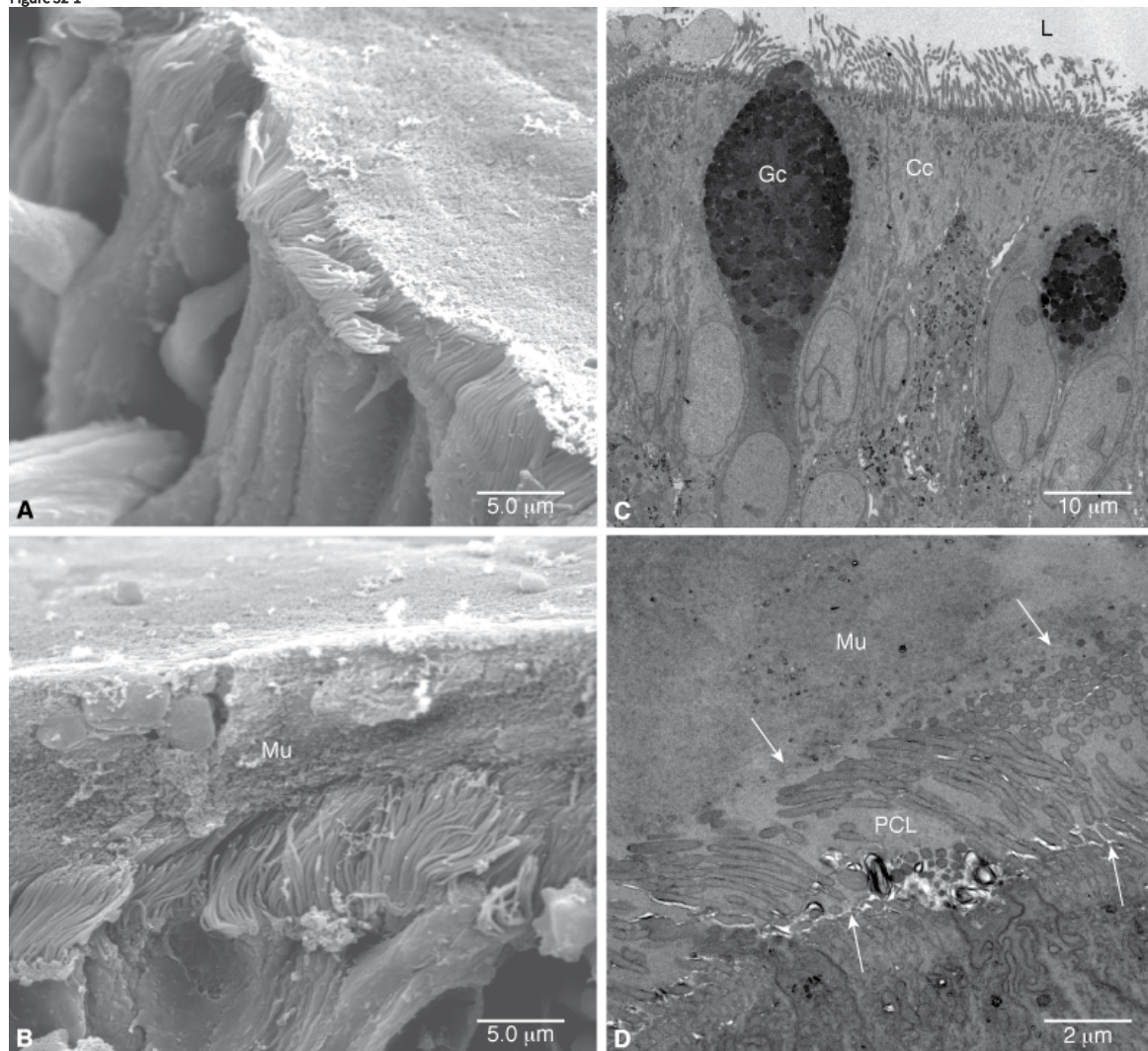
Table 52-1: Genes Coding for Secreted Mucins in the Airways

Gene	Chromosomal Locus	Tissue Distribution	Airways Predominant Distribution
MUC2	11p15	Lung, conjunctiva, middle ear, stomach, small intestine, colon, nasopharynx, prostate	Surface goblet cell Duct of submucosal glands
MUC5AC	11p15	Lung, conjunctiva, middle ear, stomach, gallbladder, nasopharynx	Surface goblet cell
MUC5B	11p15	Lung, sublingual gland, laryngeal submucosal, esophageal glands, stomach, duodenum, gallbladder, nasopharynx	Surface goblet cells Submucosal glands
MUC19	12q12	Lung, salivary gland, kidney, liver, colon, placenta, prostate	Submucosal glands

Airway Secretory Cells

The airways comprise ciliated and secretory cells, structured in a pseudostratified epithelium up to the main bronchi and as a simple cuboidal epithelium toward the peripheral airways. Mucus production in normal airways is low, approximately 10 to 100 mL/day; hence, airways are covered by a thin layer (approximately 5 μm) of mucus (Fig. 52-1B). In humans, **MUC5AC** is mainly produced by goblet cells⁷ in the proximal airways, whereas goblet cells throughout the airways and submucosal glands mostly produce **MUC5B**.⁸

Figure 52-1



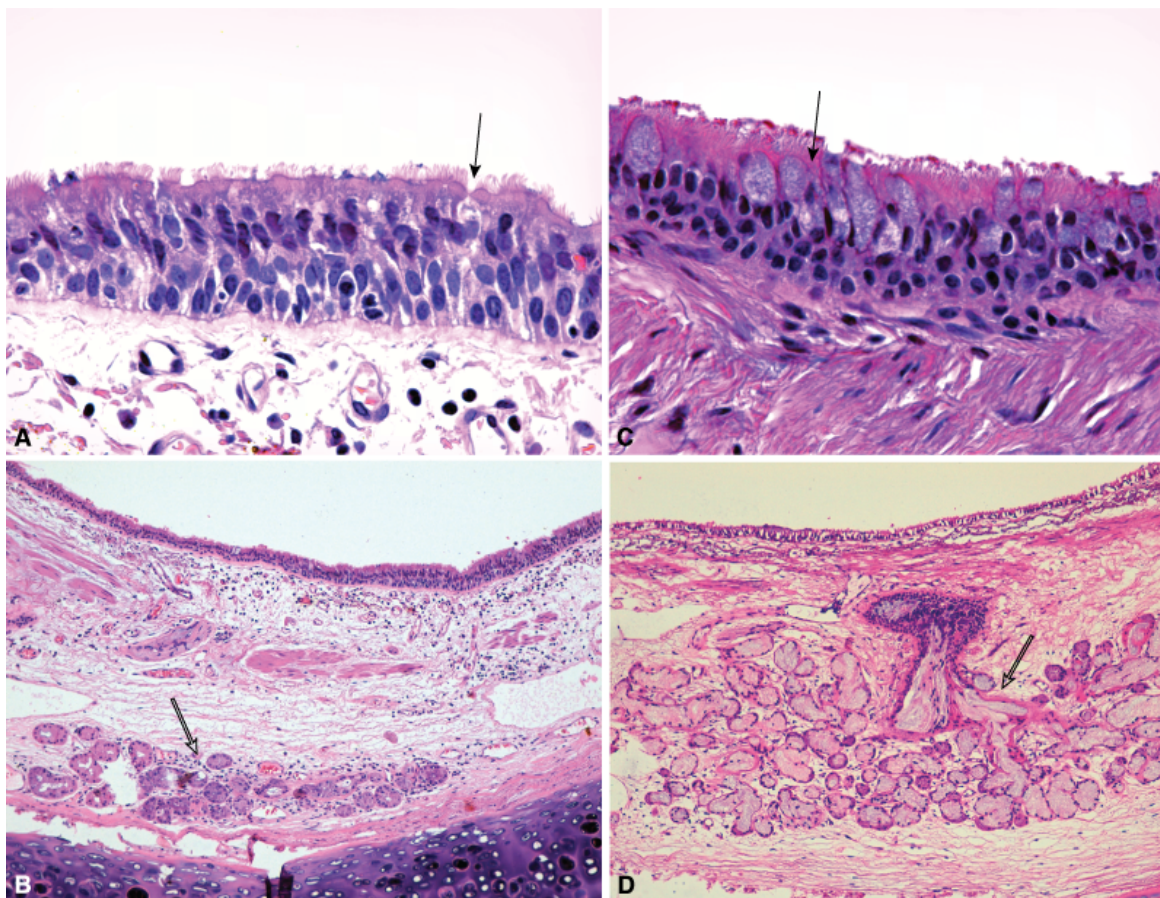
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Microanatomy of the airway epithelium. The porcine airway epithelium was fixed with perfluorocarbon/osmium and imaged with a scanning electron microscope (A and B) and transmission electron microscope (C and D). **A.** The pseudostratified tracheal epithelium and the characteristic high density of cilia on the luminal surface are shown. Rare strings of mucus are evident above the cilia (magnification $\times 3000$). **B.** A thin layer of mucus (approximately 5 μm) is demonstrated above the cilia (magnification $\times 3000$). **C.** Higher magnification of the airway epithelium comprising goblet cells (Gc) and ciliated cells (Cc). Cilia are evident in the luminal (L) surface. Of note, electron-dense mucus granules of varying size are within the goblet cell (magnification $\times 3000$). **D.** Microanatomy of the airway lining fluid. The white arrows delimitate the periciliary fluid. The electron-dense mucus layer (Mu) can be identified over the periciliary layer (magnification $\times 12,000$). (Electron micrographs courtesy Dr. Nuria Cortadells, University of Barcelona, and Laia Fernandez-Barat, Hospital Clinic, Barcelona, Spain.)

In healthy airways, epithelium goblet cells are outnumbered nearly fourfold by ciliated cells (Fig. 52-2A). Goblet cells contain cytoplasmic granules (see Fig. 52-1C), in which mucin polymers are firmly packed. When the polymers are secreted, the mucin polymers increase in size by 500 times their packed volume^{9,10} secondary to hydration.

Figure 52-2



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Representative sections of human epithelium from a healthy subject (A and B) and from a patient with chronic obstructive pulmonary disease (C and D). **A.** Magnification of the pseudostratified epithelium showing a goblet cell (*black arrow*) outnumbered by ciliated cells (magnification $\times 630$). **B.** At lower magnification, a submucosal gland is identified with no signs of dilation (*empty arrow*) (magnification $\times 100$). **C.** In patient with chronic obstructive pulmonary disease, goblet cell hyperplasia is evident. A representative goblet cell is indicated by the *black arrow* (magnification $\times 630$). **D.** Hypertrophy of submucosal gland (*empty arrow*) is shown. Reid index approximately 0.7. Additionally, a disproportionate increase in retained mucus within the gland and collecting duct is evident with resulting dilation of the structure. Hematoxylin and eosin stain. (Pictures courtesy Prof. Jose Ramirez, Hospital Clinic, Barcelona, Spain.)

In airways with an internal lumen equal to or greater than 2 mm, mucus is largely produced by submucosal glands (see Fig. 52-2B and D). Mucus produced by the gland is drained through several tubules into the main duct, which finally narrows into a ciliated duct opening into the airways.¹¹ The submucosal gland is formed by mucous cells and serous cells. Gland serous cells secrete a low viscosity compounds with antiinflammatory and antimicrobial properties. Several microbicidal products, that is, lysozyme, lactoferrin, collectins, defensins, and cathelicidins, have been identified in the airway lining fluid.^{12,13}

Regulation of Mucus Production and Secretion

A small amount of mucus is continuously produced by goblet cells and submucosal glands so as to provide a barrier overlying the airway epithelium. Mucus production can be highly increased in response to stimuli. Several aspects of MUC5AC regulation have been elucidated in detail; conversely, regulation of MUC5B expression is still not entirely understood.¹⁴ As depicted in Table 52-2, MUC5AC can be upregulated by several stimuli, among which interleukin-13 seems to play a primary role.^{15,16}

Table 52-2: MUC5AC Expression Stimulants

Inflammatory cytokines	TNF- α
	IL-1 β
	IL-4
	IL-6
	IL-9
	IL-13
	IL-17
Bacterial products	LPS
	LTA
	Peptidoglycans
	Flagellin
Growth factors	EGF
	TGF- α
	Retinoic acids
	Thyroid hormones
Pollutants	Cigarette smoke
	Acrolein
Virus	Respiratory syncytial virus
	Rhinovirus

Abbreviations: EGF, epidermal growth factor; IL, interleukin; LPS, lipopolysaccharide; LTA, lipoteichoic acid; TGF, transforming growth factor; TNF, tumor necrosis factor.

See Thai et al¹⁴ for a comprehensive review of regulation of mucins expression.

Mucus secretion is regulated by three distinct neural pathways:^{17,18} (a) the cholinergic, (b) the adrenergic, and (c) a nonadrenergic, noncholinergic (NANC) system. The cholinergic system is the most important motor control of secretion,¹⁹ and cholinergic fibers can be identified in proximity to the submucosal glands. Parasympathetic nerve stimulation causes rapid mucus release through the muscarinic M3 receptors. Adrenergic stimulation of the airway epithelium causes only a small increase in mucus production. Finally, the NANC system is identified by the mucosecretory response following adrenoceptor and cholinoceptor blockade.^{18,20} The main neurotransmitters of the NANC system are small-weight peptides, such as the vasoactive intestinal peptide, substance P, neurokinin A, and the gas **nitric oxide**.

Periciliary Liquid Layer and Mucociliary Clearance

The sol-phase, more properly termed *periciliary liquid layer*, is a thin low-viscosity fluid that extends to the height of the outstretched cilia²¹ (see Fig. 52-1D). In humans, the periciliary liquid layer is approximately 7 μ m thick and its volume is continuously regulated by the epithelial cells. Periciliary liquid layer water is driven by the luminal concentration of sodium chloride and resulting osmotic gradients. Hence, the epithelium controls the depth

of the periciliary liquid layer through purinergic signaling^{22,23} and activation of adenosine triphosphate P2Y₂ and adenosine A2b receptors. The purinergic signaling, in a paracrine–autocrine fashion, also regulates the cystic fibrosis transmembrane regulator chloride channel and the epithelial sodium channel.²⁴

Each ciliated cell has approximately 200 cilia on its surface (see Fig. 52-1A), which move, within the periciliary liquid layer, at approximately 8 to 15 Hz.²⁵ Studies report mucociliary clearance rates between 4 and 20 mm/min, depending on the method of assessment.^{26,27} Mucus clearance velocity decreases from central toward peripheral airways.^{28,29} Interestingly, early studies have hypothesized that periciliary liquid layer is largely stationary.^{30,31} More recent results^{32,33} suggest that the outermost parts of the cilia (see Fig. 52-1B) initially move the mucus layer; as a result, because of the friction between mucus and the underlying fluid, the periciliary liquid layer is dragged along and travels at a similar rate.

Mucus Clearance Via Two-Phase Gas–Liquid Transport

During breathing and cough, airflow interacts with the mucus lining the respiratory airways, which can be eventually propelled via a two-phase gas–liquid flow. Kim et al^{34–36} first studied the critical conditions necessary to move mucus through such a mechanism. They demonstrated that the main factors consisted of (a) shear stress exerted by the airflow on the liquid layer; (b) the ratio between thickness of the mucus layer and the airway diameter; and (c) the rheologic properties of secretions. Importantly, during mechanical ventilation, inspiratory and expiratory airflows exert opposite shear forces on the mucus layer. Kim et al³⁶ assessed the in vitro effects of tidal airflow, tidal volume, and respiratory rate on movement of mucus simulants and found that mucus velocity can be predicted as follows:

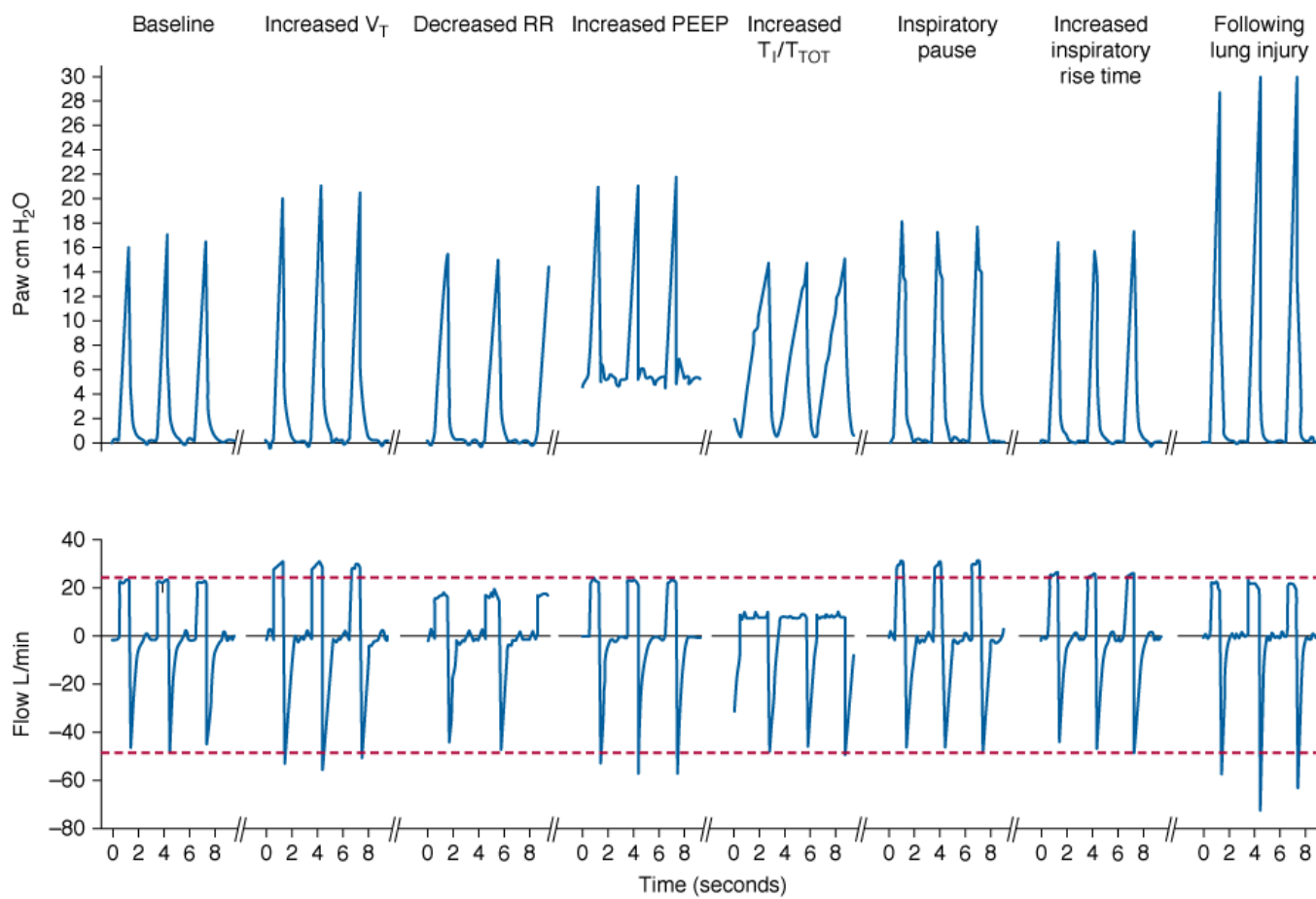
$$\text{Mucus Velocity} \sim \rho v^2 (1 - T_I/T_{TOT})^* \eta^{-1} \quad (1)$$

where ρ is the gas density, v is the absolute value of the highest airflow velocity throughout the respiratory cycle, which is inversely related to the total cross-sectional area of the airways, T_I/T_{TOT} is the duty cycle, and η is the mucus viscosity. Tidal volume and respiratory rate were not associated with mucus movement; conversely, as reported in equation (1), the absolute value of the highest airflow influenced mucus movement. Indeed, the authors found that during volume-control mechanical ventilation mucus moved in waves, and the mucus speed was not influenced by the inspiratory flow until its rate reached 90% of the expiratory flow. A recent laboratory study by Volpe et al³⁷ confirmed previous results and also emphasized the role of lung impedance on mucus clearance. Intrinsic positive end-expiratory pressure (PEEP) caused by either elevated minute ventilation or expiratory resistance respectively improved and worsened mucus clearance.

Theoretically, during mechanical ventilation the difference between inspiratory and expiratory flow rate can be modulated by adjusting ventilator settings. As clearly described in Chapter 5, lung elastance, lung volume, and airway resistances are the major determinants of expiratory flow rate during passive expiration. In contrast, inspiratory flow can be easily modulated through the ventilator. For instance, Figures 52-3 and 52-4 imply that changes in (a) mode of ventilation, (b) tidal volume, (c) respiratory rate, (d) duty cycle ($T_I/TTOT$), (e) inspiratory rise time, (f) inspiratory pause time, and (g) PEEP may all modify the inspiratory–expiratory flow rate difference. Despite consistent laboratory results, there is still great paucity of translational in vivo studies. To date, only one report in sheep has demonstrated the feasibility of mucus clearance via two-phase gas–liquid transport modulating the duty cycle.³⁸

Figure 52-3

Volume-controlled ventilation

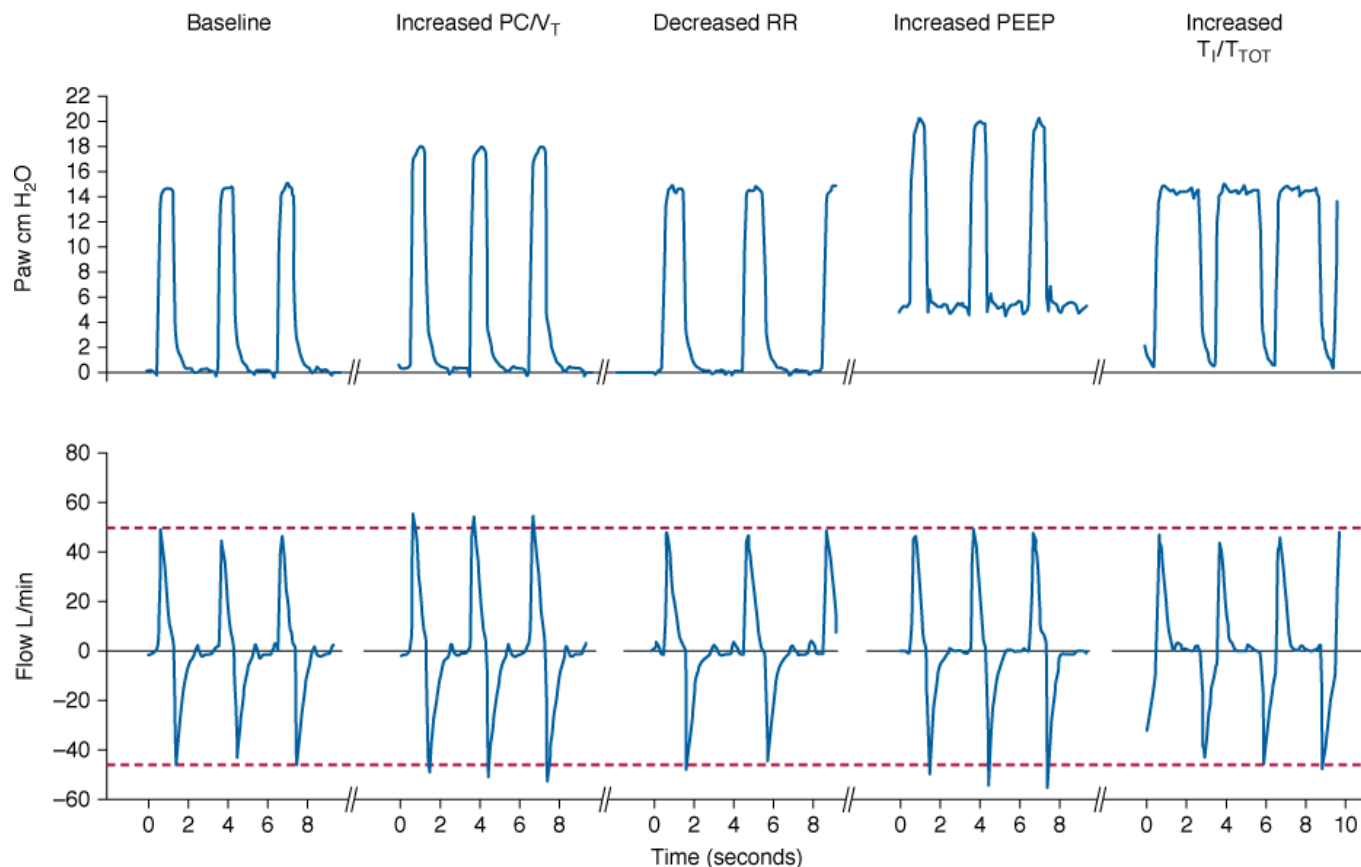


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Pressure and flow waveforms during flow-controlled, volume-cycled ventilation in a healthy pig (36 kg) obtained with a SERVO-i mechanical ventilator (Maquet, Wayne, NJ). Inspiratory and expiratory flow rates generated with baseline ventilator settings (tidal volume [V_T] 290 mL, respiratory rate [RR] 20 breaths/min, duty cycle [T_I/T_{TOT}] 0.26) are compared against different adjustments of ventilator parameters. *Dotted red lines* indicate the highest inspiratory (21.6 L/min) and expiratory (-47.2 L/min) flow rates during baseline ventilation. An increase in tidal volume to 360 mL, an inspiratory pause of 6%, and an inspiratory rise time of 5% all increased inspiratory flow. Conversely, reducing the RR to 15 breaths/min and increasing T_I/T_{TOT} to 0.75, decreased the inspiratory flow. As expected, changes in lung volume, secondary to increased tidal volume or positive end-expiratory pressure (PEEP, 5 cm H₂O), increased peak expiratory flow. Changes in elastic pulmonary conditions, induced by oleic-acid injury, caused an increase in peak expiratory flow. The effects of each individual ventilator setting on the net movement of mucus need to be thoroughly assessed by systematic in vivo studies.

Figure 52-4

Pressure-controlled ventilation



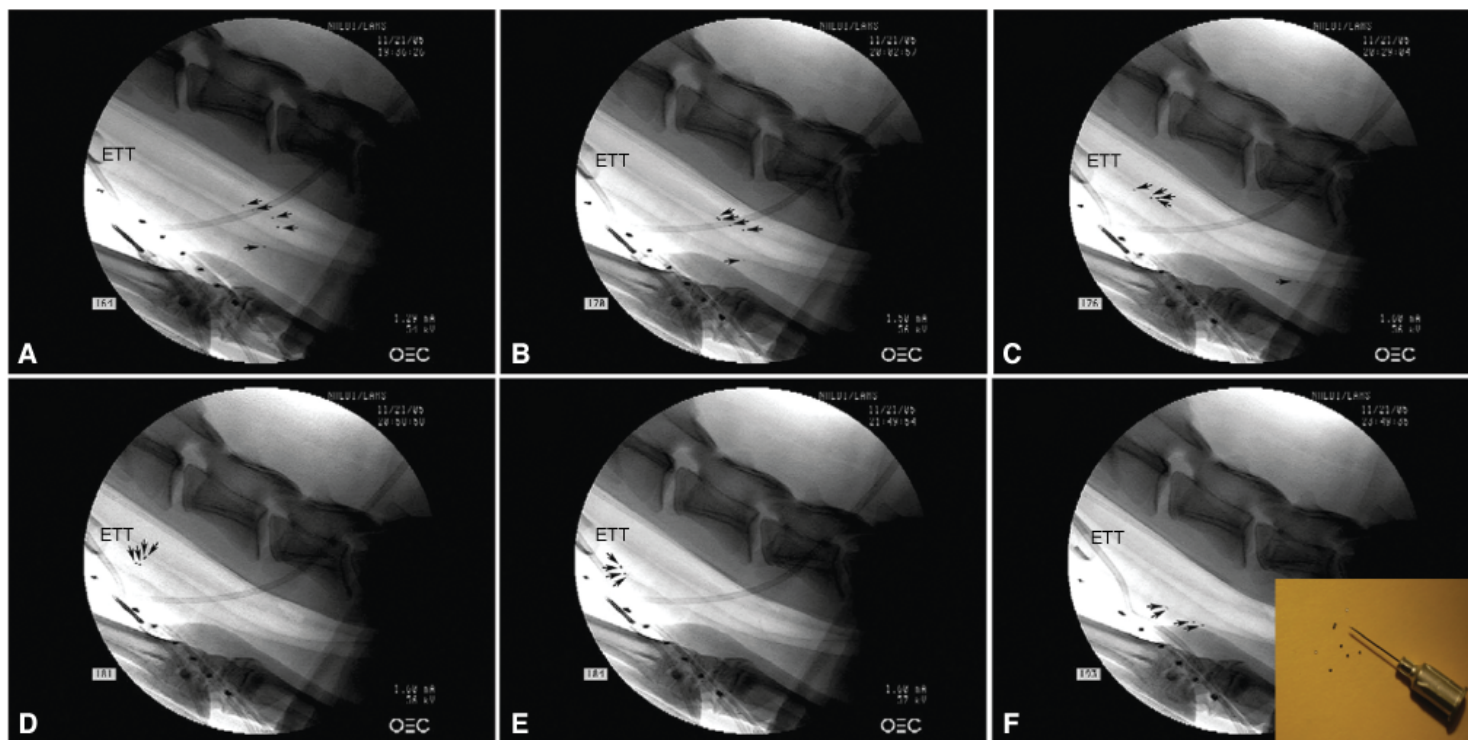
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Pressure and flow waveforms during pressure-controlled ventilation in healthy pig (36 kg) obtained with a SERVO-i mechanical ventilator (Maquet, Wayne, NJ). The effects of pressure-limited modes of ventilation on airflow rates are more difficult to predict because inspiratory flow rate is partly related to the mechanical properties of the respiratory system. Inspiratory and expiratory flow rates generated through baseline ventilator settings (pressure control [PC] 14 cm H₂O, respiratory rate [RR] 20 breaths/min, duty cycle [T_I/T_{TOT}] 0.26) are compared to various adjustments of ventilator parameters. Dotted red lines indicate the highest inspiratory (47.2 L/min) and expiratory (-47.8 L/min) flow rates during baseline ventilation. Of note, only increases in pressure control from 14 to 17 cm H₂O produced an increase in inspiratory flow. Conversely, reducing respiratory rate to 15 breaths/min and increasing T_I/T_{TOT} to 0.75 decreased the inspiratory flow. Similarly to volume-controlled ventilation, changes in lung volume secondary to increased pressure control or positive end-expiratory pressure (PEEP; 5 cm H₂O) increased peak expiratory flow.

Mechanically ventilated patients are often positioned with the trachea oriented above the horizontal (i.e., the semirecumbent position). As a result, mucus transport via two-phase gas-liquid mechanism mainly depends on a balance between the airflow shear forces on the liquid layer and gravitational force. We have studied the effects of gravity in sheep,³⁹ positioned in a model of the semirecumbent position, and found that mucus, accumulated in the proximal trachea, moved in an abnormal backward direction toward and into the lungs (Fig. 52-5).

Figure 52-5



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Tracheal mucus velocity studies in sheep after 12 hours of tracheal intubation and positioned with a tracheal orientation above horizontal. *Black/white arrows* indicate each tantalum disk, tracked to evaluate mucus transport. **A.** Following disk insufflation, five tantalum disks were deposited in the trachea; four disks were on the dorsal (nondependent) part of the trachea and one disk was on the ventral (dependent) part of the trachea (*black dots*). Fluoroscopic images taken (**B**) 24 minutes and (**C**) 59 minutes following insufflation show mucus transport toward the glottis on the nondependent part, while mucus on the dependent part of the trachea moves toward the lungs. **D** and **E.** After 80 minutes mucus almost reached the tip of the endotracheal tube, gravitated to the dependent part of the trachea, and (**F**) reversed flow back toward the lungs. The enclosed picture in F shows tantalum disks used for those experimental studies. Diameter, thickness, and weight of those disks are 0.60 mm, 0.10 mm, and 0.8 mg, respectively. (Adapted, with permission, from Li Bassi et al.³⁹)

Quantitative Assessment of Secretion Production and Removal

A quantitative assessment of mucus production and clearance in ventilated patients is essential to ascertain the benefits of novel therapeutic strategies; unfortunately, the methods for easy and reliable assessment at the bedside are still inadequate.

Mucus Production

The dynamics of mucin expression during mechanical ventilation are currently unknown and should be considered in future studies. Kirkham et al⁴ first developed a Western blot analysis that allows quantitative measurement of different mucins from retrieved mucus. In clinical settings, mucus can be easily obtained during endotracheal suctioning and stored into a mucus trap. The volume of mucus should be quantified as soon as obtained so as to avoid quantification errors secondary to dehydration. Additionally, in daily clinical practice, it is also important to report qualitative characteristics of mucus, such as color and consistency, to promptly identify any acute or developing respiratory disease.⁴⁰

Mucus Removal

The rate of mucus removal in vivo is assessed by measuring either (a) the velocity of traceable particles deposited within the trachea and/or bronchi or (b) the clearance rate of an inhaled radioactive tracer through a gamma camera. A brief description of the most common techniques are reported below and in Table 52-3, but it is important to note that most of these methods are difficult to apply in ventilated patients.

Table 52-3: Techniques for Quantitative Assessment of Mucus Clearance

	Method of Assessment	Tracer	Technical Aspects	Settings	Comments
Traceable Particles	Cinebronchofiberscopy ²⁷	Teflon disks	Disk diameter: 0.67 mm Disk thickness: 0.13 mm Disk weight: 0.13 mg 10 to 15 disks are insufflated via bronchofiberscopy Velocity is computed via the relationship of the disk size to the distance from the distal lens of the bronchofiberscope	Clinical/laboratory	Deposition of particles and frequent assessment via bronchoscope may damage the airway epithelium altering mucus transport
	Bronchofiberscopy ⁴¹	Methylene blue dye	15 µL of methylene blue dye are applied via an 18-gauge epidural catheter passed through the bronchoscope working channel The lens is positioned at the front edge of the dye and the distance from the endotracheal tube (ETT) exit measured. After a period of time, the bronchoscope is reintroduced, the leading edge of the spot relocated and the distance difference calculated to compute velocity	Clinical	Deposition of particles and frequent assessment via bronchoscope may damage the airway epithelium altering mucus transport
	Bronchofiberscopy ⁴⁵	Charcoal	5 to 15 µL of a suspension of carbon in saline The lens is positioned at the front edge of the dye and the distance from the ETT exit measured. After a period of time, the bronchoscope is reintroduced, the leading edge of the spot relocated and the distance difference calculated	Clinical	Deposition of particles and frequent assessment via bronchoscope may damage the airway epithelium altering mucus transport

	Method of Assessment	Tracer	Technical Aspects	Settings	Comments
	Fluoroscopy ⁴⁶	Teflon disks mixed with bismuth trioxide	Disk diameter: 0.64 mm Disk thickness: 0.08 mm Disk weight: 1.76 mg Timed serial lateral fluoroscopic images are taken through a C-arm fluoroscopy system to compute tracheal mucus velocity through movement of the disks A ruler with radiopaque markers is applied in close proximity of the animal's neck to help correct for magnification of fluoroscopic images	Clinical/laboratory	Fluoroscopic imaging avoids discomfort and potential bias associated with frequent bronchoscopy for assessment. Potential risks linked to radiation exposure because of frequent assessments.
	Fluoroscopy ⁴⁷	Tantalum Disk	Disk diameter: 0.60 mm Disk thickness: 0.10 mm Disk weight: 0.8 to 1 mg Timed serial lateral fluoroscopic images are taken through a C-arm fluoroscopy system to compute tracheal mucus velocity through movement of the disks A ruler with radiopaque markers is applied in close proximity of the animal's neck to help correct for magnification of fluoroscopic images	Laboratory	Fluoroscopic imaging avoids discomfort and potential bias associated with frequent bronchoscopy for assessment. Potential risks linked to radiation exposure because of frequent assessments.

	Method of Assessment	Tracer	Technical Aspects	Settings	Comments
Radioactive Tracers	Gamma camera ²⁶	Aerosolized human serum albumin microspheres with technetium-99 m	Albumin microspheres diameter: 0.5 µm Boli of radioactive microspheres are deposited on the large airways via breathing to near total lung capacity and at high airflow rate Studies may last up to 6 hours	Clinical/laboratory	The subject needs to drink water at frequent intervals to wash out any radioactivity in the esophagus Patient needs to be transferred to another department for gamma camera assessment
	Gamma camera ⁴⁹	10 to 80 µL bolus of albumin microspheres labeled with technetium-99 m	Microspheres albumin diameter: 5 to 40 µm The bolus is deposited on the tracheal mucosa through the working channel of the bronchoscope or a fine cannula	Clinical/laboratory	Patient needs to be transferred to another department for gamma camera assessment

Quantitative Assessment Via Traceable Particles

To measure the rate of mucus clearance, movement of a tracer needs to be followed over time and velocity appropriately computed. Normally, the tracer is deposited via the working channel of a bronchoscope. The anatomic location of the deposited tracer greatly influences the results, because an increased rate is reported from the peripheral toward the central airways.^{28,29} In the cinebronchofiberscopic technique, as first described by Sackner et al,²⁷ ten to fifteen Teflon disks are insufflated into the trachea and their movement continuously monitored as they approach the bronchoscope. In clinical settings, movement of several other tracers has been evaluated through bronchoscopy. In the original studies of Keller et al⁴¹ and in later reports,^{42,43} 15 µL methylene blue dye was instilled into the airways of patients undergoing anesthesia, and movement of the front edge of the dye was then sequentially measured through the bronchoscope, and velocity calculated. Charcoal markers^{44,45} have also been used. The main drawback of bronchoscopic techniques is the frequent invasive insertion of the bronchoscope, which may distort the results. The original technique was modified by Friedman et al,⁴⁶ who used fluoroscopic tracking of radiopaque disks of Teflon mixed with bismuth trioxide. In our laboratory, we use radiopaque disks made of tantalum.^{39,47} Six to eight tantalum disks are insufflated into the trachea and their velocities averaged (see Fig. 52-5). Timed, serial, lateral fluoroscopic images are taken by means of a C-arm fluoroscopy system approximately every 5 minutes to compute tracheal mucus velocity through movement of the disks.

Quantitative Assessment Via Radioactive Tracers

Inhaled γ-emitting radioactive tracers have long been used to assess mucus clearance of the whole respiratory system; to quantify the rate of mucus transport, a radioactive bolus is typically deposited in the large airways (most commonly trachea) and its movement measured through external radioactivity counters. The technique was originally described by Yeates et al²⁶ and consisted of tracheal deposition of boluses of technetium 99m (^{99m}Tc) human serum albumin microspheres (0.5 µm) suspended in isotonic droplets, achieved by breathing close to total lung capacity and at high low rates. Movement rate of those radioactive boluses was then measured through timed-serial measurement of the radioactive activity for up to 6

hours. In later reports, the radioactive bolus was deposited inside the trachea through a catheter⁴⁸ or a bronchoscope.⁴⁹ Few reports have applied these methods in tracheally intubated patients,^{50,51} mostly because the patient needs to be transferred for scintigraphic assessment, moreover, the analysis can be lengthy at times.

Mucus Dysfunctions and Source of Symptoms in Different Diseases

Patients with diseases associated with overproduction of mucus and impairment of its clearance, such as asthma, chronic obstructive pulmonary disease (COPD), and cystic fibrosis, often require mechanical ventilation. Although mechanical ventilation is lifesaving during acute worsening of those diseases, the delicate balance between mucus hypersecretion and clearance can rapidly deteriorate.

Invasive Mechanical Ventilation

Mucus retention is highly prevalent in mechanically ventilated patients because of several factors. First, following tracheal intubation, mucociliary velocity is approximately 80% slower than normal.⁵⁰ Sackner et al.⁵² demonstrated that inflation of endotracheal tube cuff lowers mucociliary velocity by 52% after only 4 hours. Second, when mucus reaches the proximal trachea outward clearance is not possible because of the inflated cuff; therefore, mucus accrues, unless it is aspirated through suctioning or enters the endotracheal tube. Third, continuous leakage⁵³ of bacteria-laden oropharyngeal secretions sustains airways infection and hypersecretion of mucus. Finally, critically ill patients are often incapable of effectively clearing retained mucus secondary to immobilization, weak cough, and muscle weakness.

Asthma

Asthma is a chronic inflammatory disease characterized by increased and inappropriate release of inflammatory mediators following antigen exposure and consequent airway smooth-muscle constriction.⁵⁴ Patients with asthma present with varying degrees of cough, shortness of breath, chest tightness, and wheezing. The role of mucus in asthma is often underestimated by clinicians⁵⁵ because mucus is typically stationary within small and medium airways and clearance is severely impaired, ultimately resulting in the inability to centralize secretions. Many mucus plugs within constricted airways may be identified in patients who die of status asthmaticus.^{56,57} Mucus in asthma is highly viscous because of an abnormal concentration of plasma proteins,^{58,59} DNA, cells, and proteoglycans. The abnormal presence of plasma proteins arises mainly from increased plasma exudation.⁶⁰

Chronic Obstructive Pulmonary Disease

COPD is highly prevalent worldwide⁶¹ and is the fourth leading cause of death according to the World Health Organization. The characteristic features of COPD are chronic bronchitis with obstruction or full closure of small airways impacted by mucus, enlargement of air spaces, and destruction of lung parenchyma resulting in loss of lung elasticity and emphysema.^{62,63} In patients with COPD, the increased mucus production is strongly associated with goblet cell metaplasia and mucus gland hypertrophy (see Fig. 52-1C and D). MUC5B is overexpressed, particularly the low-charge glycoform. Hence, the rheologic properties of mucus are altered resulting in deficient mucociliary clearance.^{64–66} Mucus hypersecretion is associated with progressive impairment of respiratory function⁶⁷ and death.⁶⁸

Cystic Fibrosis

Cystic fibrosis is caused by a defective cystic fibrosis transmembrane regulator gene,⁶⁹ which results in impaired intraluminal chloride release and increase sodium absorption.⁷⁰ The electrolytes homeostasis of the periciliary fluid is compromised, hence, the volume of the periciliary liquid layer is depleted and cilia movement is severely perturbed. Indeed, the pathologic hallmarks of the disease are viscous secretions, which adhere tenaciously to the airways and often obstruct the lumen. Although MUC5B is increased relative to MUC5AC,⁷¹ the overall concentration of those mucins is reduced,⁷² but they rapidly increase during disease exacerbations.⁷³

Role in Causing Atelectasis, Pneumonia, and Other Disorders

When mucus is retained, several complications occur as a consequence of (a) progressive reduction of the airways lumen, impacted by mucus, and/or (b) overgrowth of pathogens entrapped within the retained mucus. Excessive accumulation of secretions within the respiratory system and artificial airways makes it difficult to exhale, resulting in intrinsic PEEP and increased work of breathing. Shah et al.⁷⁴ found a significant reduction in the internal volume of the endotracheal tube, which was proportional to the duration of mechanical ventilation. Indeed clinicians, upon extubation, often acknowledge the vast presence of secretion within the endotracheal tube. The mucus-related increase in airflow resistance and work of breathing⁷⁵ causes breathing discomfort and may retard ventilator weaning.⁷⁶

Mucus retained in the peripheral airways can ultimately cause atelectasis and gas-exchange impairment. In intubated patients, mucus can be suctioned only in the larger airways; hence, mucus within the small airways may produce complete and irreversible closure, because the pressures necessary to reopen small airways are extremely high and difficult to achieve safely in a ventilated patient. Patients at particular risk of developing mucus-related atelectasis are those with asthma, COPD, cystic fibrosis, infants, children, and those undergoing surgery. In surgical patients, mucociliary clearance becomes impaired in the operative room,⁷⁷ through the use of anesthetics and tracheal intubation; moreover, inefficient cough, associated with the underlying diseases or the anatomical site of the surgical procedure, increase the risk.

Retained mucus is also associated with lung infections. Patients with underlying respiratory diseases experience periodic exacerbations caused by pulmonary infections. The microbicidal products contained in mucus exert a time-limited effectiveness; Cole et al⁷⁸ elegantly demonstrated that bacteria added to nasal mucus rapidly acquired resistance to the antimicrobial factors within 24 hours. Those findings suggest that the antimicrobial properties of mucus and clearance work synergistically, and retained mucus eventually become colonized.

Airway Suctioning

In intubated or tracheostomized patients, endotracheal suctioning consists of aspirating secretions retained within the artificial airway, trachea, and main bronchi via a suction catheter. Suctioning is one of the most common daily procedures. It should be performed only by qualified personnel, namely, physicians, respiratory therapists, or nurses, with appropriate skills and training to recognize the need for suctioning, perform it appropriately, and promptly respond to any potential complication.

Indications and Technical Procedure

Indications

The primary indication for endotracheal suctioning is to maintain patency of the airways, but it is also commonly performed to obtain secretions for diagnostic purposes. Current guidelines recommend that suctioning should not be performed routinely⁷⁹ to minimize the risk of unnecessary complications.

Several parameters have been suggested to detect the need for suctioning (Fig. 52-6). Researchers have focused especially on the typical sawtooth pattern of airflow in patients with secretions (Fig. 52-7). Jubran and Tobin⁸⁰ first proposed that retained mucus could be detected via the flow-volume loop, and reported that a sawtooth pattern had a good sensitivity and specificity. Guglielmotti et al⁸¹ subsequently confirmed that the best predictors of the need of suctioning were (a) respiratory sounds heard via a stethoscope over the trachea, and (b) the sawtooth pattern on a real-time flow-volume loop (sensitivity 0.82, specificity 0.70).

Figure 52-6

POTENTIAL INDICATORS FOR ENDOTRACHEAL SUCTIONING:

- 1) Visible secretions within the artificial airway
- 2) Dyspnea or apparent increased work of breathing
- 3) Pathological sounds (i.e., rhonchi, coarse and gurgling) heard *via* a stethoscope over the trachea
- 4) Progressive pulse oxymetry desaturation not otherwise explained
- 5) Increased peak inspiratory pressure in volume-controlled MV or decreased V_T in pressure-controlled MV
- 6) Sawtooth pattern identified at the real-time flow-volume loop

**PREPARATION PRIOR ENDOTRACHEAL SUCTIONING:****EQUIPMENT**

- 1) Consider closed suctioning system (or swivel connector) in patients with severe respiratory disease requiring high level of PEEP and FiO_2
- 2) Ratio of catheter OD/artificial airway ID should be ≤ 0.7 in adults and ≤ 0.5 in children
Suction catheter diameter (Fr) = Artificial airway internal diameter $\times 3/2$
- 3) Choose catheters with nonparallel lateral suction holes. In selected cases, a curved tipped catheter may be helpful to suction mucus within bronchi difficult to access
- 4) Prepare suction tubing
- 5) The level of vacuum should always be adjusted to the lowest level of vacuum that efficiently removes secretions. In general, vacuum ≤ 150 mm Hg in adults and 80–100 mm Hg in infants.

PATIENT PREPARATION

- 1) In patient on MV, heart rate, pulse oximetry, arterial pressure, and in particular cases, intracranial pressure, should be monitored throughout the procedure
- 2) If patient is conscious, explain the procedure
- 3) The use of saline to hydrate inspissated secretions and centralize secretions through cough lacks of clear evidence. If secretions appear dehydrated, check if humidification of inspired gases is appropriate
- 4) Hyperoxygenate for 120 sec using in adults FiO_2 of 100% and a 10% increase from baseline in infants. Of note, several ventilators now provide a single operational control for suction support

OPERATOR PREPARATION

- 1) Wash hands
- 2) Wear disposable apron and protector visor, particularly during open suctioning
- 3) Wear sterile glove on the hand manipulating the catheter, and nonsterile glove on the other hand
- 4) During open suctioning, retrieve the catheter from the sleeve and hold it. In particular, avoid manipulation of the distal catheter tip
- 5) Connect the catheter to suction tubing.
- 6) During closed or quasi-closed suctioning, the ventilator should be allowed to trigger to provide gas in response to negative pressure

**PROCEDURE:**

- 1) Insert the catheter into the artificial airway and advance its distal tip up to the proximal trachea
- 2) Intermittent suction should be applied through occlusion of the catheter suction control port
- 3) Withdraw suction catheter continuously rotating it
- 4) The procedure should last 15 sec or less from catheter insertion
- 5) Check and appropriately treat any potential cardiovascular, pulmonary, and neurologic complication throughout the procedure

**FOLLOW-UP:**

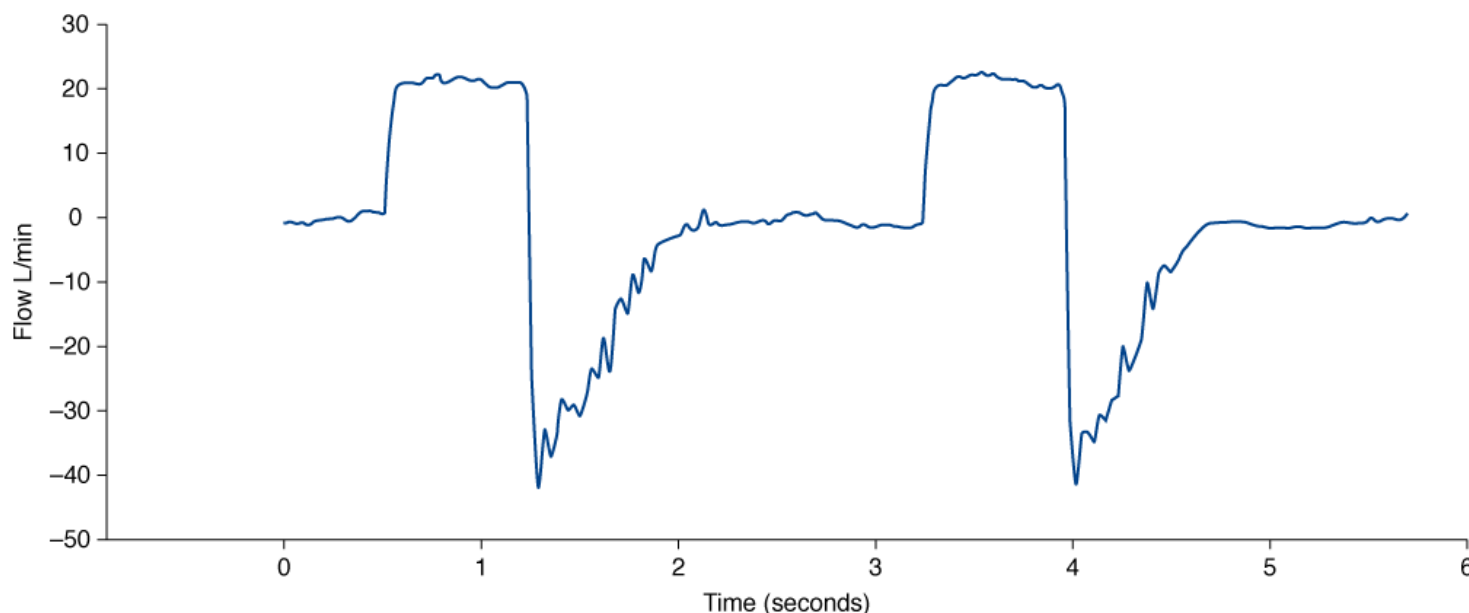
- 1) Assess respiratory cardiovascular and neurologic parameters
- 2) Consider hyperoxygenation
- 2) Consider recruitment maneuver in selected patients with ARDS and ALI
- 3) Rinse suction tubing and, in case of closed suctioning system, suction catheter
- 4) Dispose of the suction catheter apron and gloves into the proper containers

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Suggested operative protocol for endotracheal suctioning. *ARDS*, acute respiratory distress syndrome; *ALI*, acute lung injury; FiO_2 , inspiratory fraction of oxygen; *ID*, internal diameter; *MV*, mechanical ventilation; *OD*, outer diameter; *PEEP*, positive end-expiratory pressure; V_T , tidal volume.

Figure 52-7



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Flow waveform during volume-controlled mechanical ventilation. The typical expiratory sawtooth pattern used as an indicator for endotracheal suctioning can be recognized. Of note, the presence of water condensation within the circuit should be excluded before proceeding with suction.

Endotracheal Suctioning Procedure

Endotracheal suctioning begins with inserting a suction catheter into the artificial airway. Because there is no obvious advantage with deep suctioning versus shallow suctioning, the tip of the catheter should be placed in close proximity to the tip of the endotracheal tube before applying vacuum. As the catheter is pulled back, intermittent suction should be applied and the catheter continuously rotated. Shallow suctioning is particularly advisable in infants to minimize complications.^{82,83} The only goal of suctioning is aspiration of secretions; hence, application of negative pressure, when the catheter is not close to secretions, or when all the mucus has already been aspirated (i.e., prolonged suctioning), only increases the risk of complications with no improvement in efficacy. Suctioning should last no more than 15 seconds⁷⁹ after inserting a catheter into an artificial airway.

Selection of Suction Methods and Catheters

Open versus Closed Suctioning

Endotracheal suctioning can be performed either by disconnecting the patient from the ventilator (open suctioning) or through use of a suction catheter that does not require ventilator disconnection (closed suctioning), which is located inside a plastic contamination shield and in-line with the ventilator circuit.^{84,85} Closed suctioning is beneficial in patients who require high levels of PEEP and inspired fractional concentrations of oxygen (FiO_2). Several studies in adults with acute lung injury and acute respiratory distress syndrome (ARDS) have confirmed that, unlike open suctioning, closed suctioning prevents reduction of lung volume and desaturation.^{86–88} These studies suggest that closed suctioning is especially appropriate in patients with most severe lung failure (partial pressure of arterial oxygen [$\text{PaO}_2/\text{FiO}_2 \leq 200$]), who are intolerant of even brief disconnections from the ventilator. In studies by Cereda et al⁸⁶ and Maggiore et al,⁸⁸ open suctioning caused almost 1.5 L volume loss, half of which was secondary to the ventilator disconnection.⁸⁸ Results of studies assessing open and closed suctioning in newborns with respiratory distress syndrome^{89–92} are consistent with data from adults.

Closed suctioning is also been introduced to reduce environmental contamination,^{93,94} exogenous colonization of the airways via suction catheters⁹⁵ and potentially ventilator-associated pneumonia. Three meta-analyses⁹⁸ have compared closed and open suctioning and found no benefits in preventing ventilator-associated pneumonia. Siempos et al⁹⁷ analyzed data from nine randomized trials, comprising of 1292 patients; data pooled from four showed a higher incidence of respiratory tract colonization using closed suctioning.

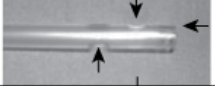
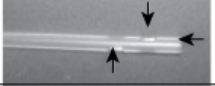
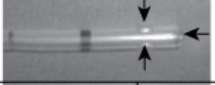
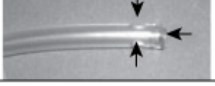
During closed suctioning it is important to note that triggered ventilator autocycling partially compensates for the loss of pressure, and investigators^{99,100} have shown that the distal dispersion of secretions by ventilator airflow may reduce the effectiveness of closed versus open suctioning.

Catheter Characteristics

The main features that should be considered when choosing the most appropriate suction catheter are (a) the ratio between catheter outer diameter of the catheter and the internal diameter of the artificial airway; and (b) the size, number, and location of the suction holes. The outer diameter of the suction catheter greatly influences the loss of lung volume during the procedure. A larger catheter increases endotracheal tube resistance and facilitates loss of lung volume, because aspirated gas is not rapidly replaced by gas flowing through the endotracheal tube. These theories were clearly elucidated in early reports by Rosen et al^{101,102} and confirmed in laboratory studies assessing pediatric^{103,104} and adult¹⁰⁵ catheters. Evidence from those studies strongly suggests that the ratio of suction catheter outer diameter to artificial airway internal diameter should be 0.7 or less in adults and approximately 0.5 in children.

Several investigators have evaluated intrinsic characteristics of the suction catheter to identify factors associated with efficacy and safety.^{106–109} Sackner et al¹¹⁰ first demonstrated that invagination of the tracheal mucosa into the side holes frequently occurred with all the different catheter designs. In a more recent report, Shah et al¹⁰⁹ assessed six suction catheters (Fig. 52-8) with different designs and found that all performed equally with low viscosity mucus simulant; when viscosity resembled airway secretions, larger and nonparallel side holes were more efficient. In certain instances, a curved tipped catheter^{111–113} may help when suctioning mucus within bronchi difficult to access with a straight catheter.

Figure 52-8

Catheter	Side-Hole Diameter	Size	Whistle Design	Side-Hole Design
Cardinal Tri-Flo	3 mm	14 Fr		
Kendall Regu-vac	4 mm, 6.35 mm apart	14 Fr		
Kendall Sensi-vac	5 mm, 9.52 mm apart	14 Fr		
Medline Delee	4 mm, 9.52 mm apart	14 Fr		
Medline Whistle Tip	4 mm	14 Fr		
Portex suction tray	4 mm	14 Fr		

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14 Fr suction catheters studied in vitro by Shah et al employing a mucus simulant with viscoelastic properties similar to human mucus. Nonparallel position and larger diameter of the suction holes were the major determinants of catheter's efficacy. (Used, with permission, from Shah et al.¹⁰⁹)

Additional Equipment

In ventilated patients, strict monitoring during endotracheal suctioning is strongly advised.⁷⁹ It is especially important to monitor heart rate, oxygen saturation, arterial pressure, and, in particular cases, intracranial pressure, throughout the procedure.

Vacuum Adjustable Regulator

The level of vacuum should always be adjusted before suctioning, so as to improve efficacy and minimize complications. Few studies have evaluated the efficacy and safety of suctioning at increasing vacuum levels. The latest guidelines⁷⁹ recommend a vacuum no higher than 150 mm Hg in adults and 80 to 100 mm Hg in neonates, although the lowest level of negative pressure that efficiently removes secretions should always be favored.

Additional Equipment for Sterile Suctioning

Theoretically, during open suctioning there is a risk for environmental contamination from the patient and exogenous colonization of the patient. First and foremost, hand hygiene before and after the procedure is mandatory. Endotracheal suctioning is an invasive procedure and the use of sterile gloves is advisable, although there are no clinical studies specifically investigating the association between the use of unsterile gloves and respiratory

infections. Blackwood et al¹¹⁴ reported that gloves are important even during closed suctioning because of potential contamination of staff hands from the flush port.

Swivel Connector

A quasiclosed suctioning system, via a swivel adapter¹¹⁵ positioned at the proximal tip of the endotracheal tube, is an alternative approach to minimize the loss of lung volume and desaturation during suctioning.^{88,116} Investigators^{87,88} found that suctioning through either a swivel adapter or a closed suctioning system similarly limited the decrease in lung volume as compared with open suctioning. Importantly, as during closed suctioning,¹¹⁷ the ventilator should be allowed to autcycle to compensate for the loss of gas.

Sterile Water

The use of 2 to 5 mL boluses of saline, instilled into the endotracheal tube before suctioning to induce cough and hydrate inspissated secretions remains controversial and lacks clear evidence. The first concern is that saline instillation, particularly in the semirecumbent position, may drip into the lungs and cause serious adverse effects, such as oxygen desaturation,^{118,119} tachycardia,^{119,120} dyspnea, and anxiety.¹²¹ Second, several studies^{122,123} have consistently demonstrated the presence of bacterial biofilm within the endotracheal tube, hence, instillation of saline may dislodge pathogens.¹²⁴ Caruso et al¹²⁵ published a report on 262 patients randomized to receive either isotonic saline instillation before suctioning or no treatment. The authors found a lower incidence of ventilator-associated pneumonia, and no significant differences in secondary outcomes. Because of limited evidence and the potential for adverse effects, routine saline instillation cannot be recommended; moreover, when mucus progressively becomes thicker, the humidifier's performance should be checked.

Complications of Endotracheal Suctioning and Preventive Measures

After discharge from an intensive care unit, patients remember endotracheal suctioning as one of the most unpleasant and painful procedures,^{126–128} such memories may affect later quality of life.¹²⁹

Once inside the trachea, the suction catheter frequently adheres to the tracheal mucosa, and may lead to the impairment of mucociliary transport¹³⁰ and severe mucosal injury.^{110,131} Sackner et al¹¹⁰ first addressed this problem by devising a novel suction catheter with a modified tip to prevent contact against the tracheal mucosa. In a later report¹⁰⁶ the new catheter was compared against standard catheters in dogs, and the investigators demonstrated that the risk of mucosal injury was specifically associated with deep suctioning rather than catheter design or vacuum level. Intermittent suctioning is recommended to prevent mucosal invagination into the suction holes.

Endotracheal suctioning is associated with several cardiovascular, pulmonary, and neurologic complications. Bradycardia and hypertension are the most frequent cardiovascular complications, mostly as a result of a vagal reflex and resulting in increases in parasympathetic and sympathetic activities.^{132–135}

Hypoxemia during suctioning is extremely common, particularly in adults with severe lung disease^{86,88} or infants.⁹² As described above, closed or quasiclosed systems have been used to prevent those complications. Hyperoxygenation before the procedure is also used to prevent hypoxia,^{136,137} and several ventilators now provide a control for automatic preoxygenation and post-oxygenation. A meta-analysis,¹³⁸ based on pooled data from fifteen studies in adults, demonstrated a 49% reduction of suction-related hypoxia when pre-hyperoxygenation and post-hyperoxygenation was applied. There is no clear evidence as to the level of FiO_2 that should be used to hyperoxygenate. In adults, FiO_2 of 100% is widely used. Hyperoxia, however, may be deleterious,¹³⁹ particularly in neonates; hence, in these instances, a 10% increase^{79,140} from baseline is recommended. A recruitment maneuver applied after suctioning in selected patients with ARDS or acute lung injury is associated with more rapid return of baseline lung volume and reversal of hypoxia.^{88,141,142} Finally, severe bronchoconstriction has been demonstrated during suctioning in animals^{143,144} and humans.¹⁴⁵

In patients with intracranial hypertension, an abrupt burst of coughing provoked by tracheal stimulation could increase the intracranial pressure; if there is not a concomitant increase in cerebral blood flow, brain ischemia may ultimately result.^{146–148} Several drugs have been tested, mainly in adults, to blunt the responses to suctioning, including barbiturates,¹⁴⁹ narcotics,^{150,151} anesthetic agents,^{152–155} and neuromuscular blocking agents.^{149,156,157} These drugs should be indicated on a case-by-case basis and the potential adverse hemodynamic effects of such agents^{151,158} should be balanced against the perceived benefits.

Lastly, suctioning may increase endogenous and exogenous colonization of the lower airways. Manipulation of the suction catheter before insertion,¹⁵⁹ passage of the catheter within the colonized artificial airway,¹⁶⁰ and mucosal injury caused by negative pressure¹¹⁰ are possible culprits involved in suctioning-related colonization of the airways. In ventilated newborns, transient bacteremia has been associated with endotracheal suctioning.¹⁶¹ Finally, a number of studies found that closed suctioning catheters were easily colonized and associated with higher incidence of airways colonization;^{161–165} nevertheless, the evidence is still controversial and, as described above, no increase in incidence of ventilator-associated pneumonia was found, even when the catheter was not changed daily.¹⁶⁶

Manual Ventilation Techniques

Exhaled Ventilation Techniques

In ventilated patients, who have excessive mucus production but are unable to produce an efficient cough, physiotherapists often apply manual and mechanical procedures with a goal of increasing expiratory flow and facilitating outward drainage of secretions¹⁶⁷ via annular two-phase gas–liquid flow.³⁶

Manual hyperinflation to promote mucus clearance was originally described by Clement and Hubsch in 1968.¹⁶⁸ Manual hyperinflation is performed by disconnecting the patient from the ventilator and providing, via a resuscitator bag, a slow inspiration of a larger tidal volume, an inspiratory hold, and a quick release of circuitry pressure to assure a high expiratory flow rate. Extreme heterogeneity exists for the procedure and equipment.¹⁶⁹ A recent Dutch survey¹⁷⁰ substantiates this and relates how even the most important features of the procedure are frequently overlooked. Manual hyperinflation is commonly applied for mobilization of retained secretions, to recruit atelectatic lung regions, and to improve gas exchanges. During the procedure, the lack of respiratory monitoring and the potential to reach very high airway pressure¹⁷¹ are of great concern. Significant hemodynamic fluctuations during the procedure have been reported.¹⁷² The procedure should be performed by caregivers trained in the maneuver¹⁷³ and is usually contraindicated in patients with hemodynamic instability, undrained pneumothorax, intracranial hypertension, severe bronchospasm, and ARDS requiring a high level of PEEP or FiO_2 .¹⁷⁴ Among studies that specifically looked at mucus transport during manual hyperinflation, Jones et al¹⁷⁵ assessed transport of simulated mucus of three different viscosities and found that outward movement of mucus was consistently increased, particularly low viscosity artificial mucus. In eighteen intensive care unit patients positioned on their side, Hodgson et al¹⁷⁶ found increased retrieval of mucus following manual hyperinflation. One report¹⁷⁷ demonstrated even greater efficacy when the procedure was combined with the lateral Trendelenburg position.

It is commonly advocated that tidal volume be increased to 50% above baseline during manual hyperinflation, or an inspiratory pressure lower than 40 cm H₂O be achieved. Singer et al,¹⁷² however, found that manual hyperinflation generated heterogeneous ranges of peak inspiratory pressure and tidal volume; therefore, inline manometers, spirometers, or both should be used both in research and clinical setting.

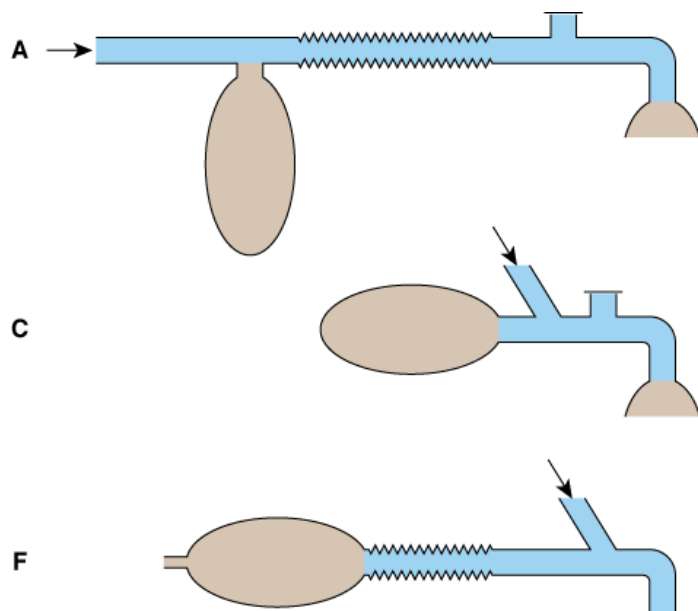
Three studies^{178–180} compared the effects of manual hyperinflation against hyperinflation delivered by the ventilator; amounts of aspirated secretions were similar.^{178,179}

Ventilatory Circuits for Manual Hyperinflation

The type of circuit used during manual hyperinflation strongly influences the resulting airflow rates, airway pressures, and mucus clearance. Circuits can be classified on the basis of the characteristics of the inflation bag, location of the fresh gas inlet, and expiratory valve.

Flow-Inflating Bag Circuits

The most common rebreathing circuits used for manual hyperinflation are the Mapleson-A/Magill, Mapleson-C, and Mapleson-F (Fig. 52-9). The circuits consist of a highly compliant 2 L rubber or neoprene bag (with an open expiratory port in the Mapleson-F circuit) and an adjustable pressure-limiting valve (Magill and Mapleson-C circuits). The compliant bag allows the operator to better assess the patient's respiratory compliance during the procedure. The gas inlet is close to the patient in the Magill and Mapleson-C circuits and it is commonly connected to 100% oxygen at a flow rate of 15 L/min. During inspiration, an appropriate adjustment of the adjustable pressure limiting valve is required to inflate the bag and to provide ventilation at the necessary respiratory rate and volume. Differently, to ensure a rapid exhalation, the bag is fully and promptly released and the adjustable pressure limiting valve can be adjusted to the fully open position, and manually held closed during inspiration.



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Most common semiclosed ventilatory circuits for manual hyperinflation delivered with a flow-inflating bag. The Mapleson A/Magill, Mapleson-C, and Mapleson-F are shown. Circuits differ in terms of location of fresh-gas inlet and presence of expiratory port (F) or valve (A and C). *Black arrows* indicate the fresh gas inlet.

Self-Inflating Bag Circuits

The self-inflating bag circuits comprise a silicone bag that is connected to the patient's artificial airway via a non-rebreathing valve assembly. At the other side of the bag are located the fresh-gas inlet and the oxygen-reservoir bag. Several studies in the recent decades have assessed performance of the Laerdal 1.6 L adult bag (Laerdal Medical Corp, New York, NY) and the Air Viva bag (now discontinued).

The ratio between the inspiratory and expiratory flow rates is the most important factor during manual hyperinflation. As expected, a longer inspiratory time is associated with a slower inspiratory flow rate.¹⁸¹ In contrast, expiratory flow depends on (a) nonmodifiable factors, such as airways or endotracheal tube resistances and respiratory compliance, and (b) modifiable factors, that is, end-inspiratory lung volume and breathing circuits. In vitro studies^{181,182} have compared performance of different breathing circuits for manual hyperinflation and have shown that the Mapleson-C and Mapleson-F circuits generate faster expiratory flow than do circuits with self-inflating bags. Moreover a clinical study by Hodgson et al¹⁸³ of patients receiving manual hyperinflation with a Laerdal and Mapleson-C circuit, found more secretions retrieved with the latter. In a more recent in vitro study,¹⁸⁴ the Mapleson-C circuit generated greater peak expiratory flows when compared to the Mapleson-A circuit, probably because of intrinsic characteristics of the exhalation valve.

Role of Pharmacologic Agents

Numerous drugs commonly used in ventilated patients profoundly affect mucociliary clearance and increase mucus retention. Oxygen,¹⁸⁵ inhaled anesthetics,^{42,43,186} and narcotics¹⁸⁷ have shown dose-dependent and time-dependent inhibitory effects on mucociliary clearance. Because mucus is composed mostly of water, administration of fluids and diuretics may affect hydration and rheologic properties of airway mucus.¹⁸⁸ Furosemide has a direct inhibitory effect on the NaK(Cl)₂ cotransporter located in the basolateral membrane of the airways epithelium and may influence the volume of the periciliary liquid layer. A study in mechanically ventilated patients¹⁸⁹ revealed some deleterious effects of furosemide on mucus transportability; further research is needed particularly because of conflicting laboratory results.^{190,191}

Mucoactive Agents

Remarkably little data exist on the use of mucoactive agents in ventilated patients, and most of the knowledge in the field is provided by clinical trials assessing long-term effects of such agents in patients with underlying chronic respiratory diseases. Mucoactive drugs used in intensive care unit settings can be divided in mucolytics and mucoregulators.

Mucolytics alter viscoelastic properties of mucus by either dissociating the mucus polymers or the cellular residuals within mucus. *N*-acetylcysteine dissociates the disulfide bonds that generate the final mucin polymeric structure, hence, enhances fluidity of mucus to facilitate its clearance.¹⁹² *N*-acetylcysteine can be administered intravenously, orally, and aerosolized, although it can cause airway irritation when inhaled.¹⁹³ Only one randomized, controlled study¹⁹⁴ has been conducted in ventilated patients. *N*-acetylcysteine had no significant benefit on viscosity of secretions or the risk of mucus plugs. Likewise, sodium 2-mercaptoethane sulphonate (MESNA) is a mucolytics agent¹⁹⁵ that reduces mucus disulfide bonds. The agent can be administered via a nebulizer¹⁹⁶ or instilled directly into the airways.¹⁹⁷ MESNA may induce severe bronchospasm and coadministration of a bronchodilator is advisable in intubated critically ill patients. Increased mucus concentration of highly polymerized DNA, caused by degenerated polymorphonuclear leukocytes, greatly alters the viscoelastic properties of mucus. The resulting thick and tenacious mucus narrows the airways and may cause atelectasis. Recombinant human deoxyribonuclease hydrolyses DNA and has been largely used in patients with cystic fibrosis.¹⁹⁸ Scattered work has been published in ventilated pediatric patients, on the use of recombinant human deoxyribonuclease, inhaled or administered via bronchoscopy, to resolve established atelectasis secondary to mucus plugs.^{199–205}

Mucoregulators affect mucus production and secretion. Among these drugs, anticholinergics, glucocorticoids, and β_2 -adrenergic agonists are routinely prescribed in critical care, mainly to reduce bronchospasm and airway inflammation, often overlooking the important effects on airways secretions. Inhaled anticholinergics agents, such as *atropine*, *ipratropium*, and *tiotropium*, thwart mucus hypersecretion of submucosal glands. Glucocorticoids indirectly influence production of mucus by reducing airways inflammation. Similarly, macrolide antibiotics exert an immunomodulatory effect that can result in decreased mucus production. β_2 -adrenergic agonists enhance mucociliary clearance.^{206–209} The bronchodilator action may also favorably influence the peripheral interaction between airflow and retained mucus.

Important Unknowns

As already discussed, most of the quantitative techniques to assess mucus clearance were developed approximately 40 years ago, and currently they are applied in laboratory settings. Thus, surrogate end points have been used in research, that is, changes in oxygenation, pulmonary mechanics, and amount of aspirated secretions—all of which increase the odds for negative studies²¹⁰ and produce controversy in the field.

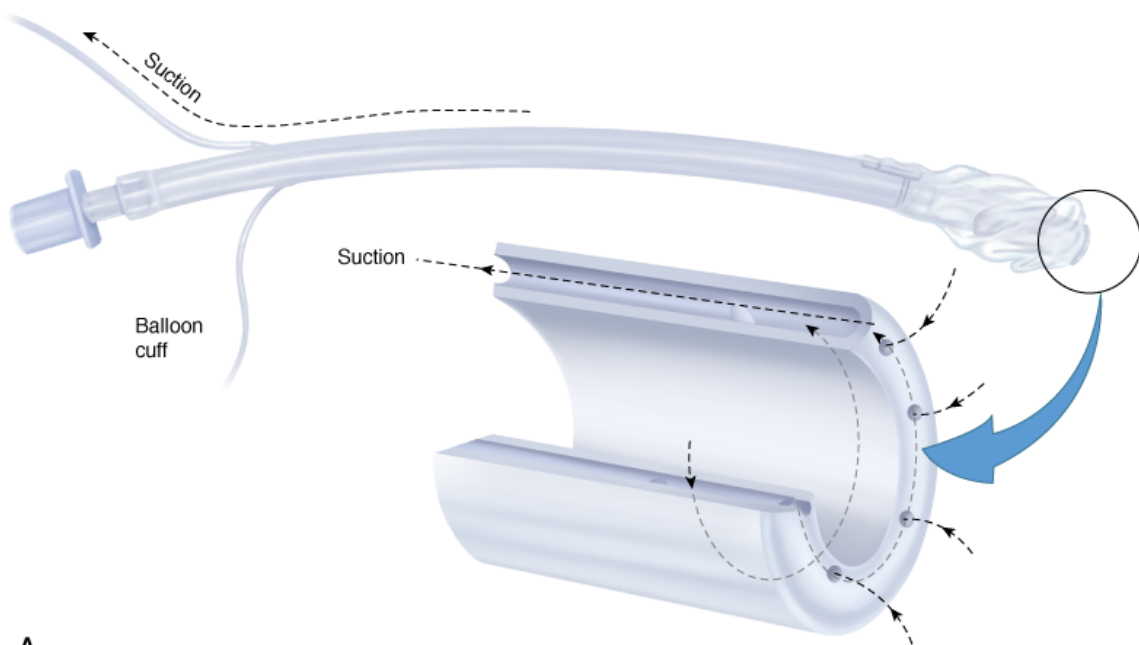
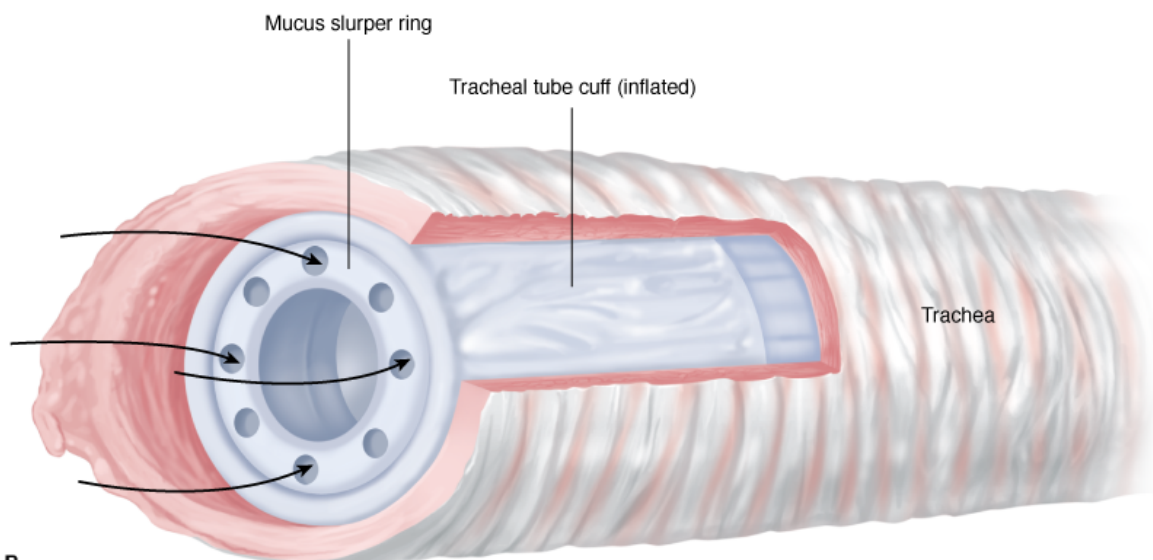
First and foremost, in daily clinical practice, indications for endotracheal suctioning remain uncertain. Studies have demonstrated that respiratory waveforms can predict the presence of retained secretions,^{80,81} yet we do not know whether it is safe to allow mucus to accumulate in the proximal trachea until it is detected by such monitoring. The proximal trachea is often colonized by seepage of bacteria-laden oropharyngeal fluids; hence, mucus may provide a perfect media for exponential growth of pathogens that eventually flow³⁹ backward into the lungs driven by gravity. Second, several laboratory^{36,37} and animal studies^{38,39} suggest that the net movement of retained mucus is governed by gravity and tidal airflow forces. Those theories have been extrapolated to the pathogenesis of ARDS because mucus and noxious fluids, from an initially contained locus of infection, may propagate through the airways and ultimately cause ARDS.^{211,212} Finally, although hypersecretion of mucus is universally found during prolonged ventilation, even in patients without underlying respiratory disease, the pathophysiologic basis of such hypersecretion is still unknown and there are no studies assessing the effects of mechanical ventilation on expression of mucins.

The Future

Several promising areas exist for future laboratory and clinical research. Development of new reliable techniques to quantitatively assess movement of airway secretions within the proximal and lower airways is the first priority. Graf et al²¹³ have employed sequential computed tomography to trace a radiopaque mucus simulant in preserved ex vivo porcine lungs. Although the technique was designed to track lung biofluids, the experimental approach provides an attractive option for future studies.

Given the adverse effects and the frequent use of endotracheal suctioning, efforts should be made to devise new strategies to clear mucus from the airways more efficiently and safely. Two prototypes have been described. The Mucus Slurper²¹⁴ (Fig. 52-10) consists of a novel endotracheal tube that provides intermittent and automatic suction at its tip. Using the Mucus Slurper, we were able to avoid endotracheal suctioning over 72 hours of mechanical ventilation, and we prevented any accumulation of mucus within the endotracheal tube and proximal trachea.²¹⁵ The Mucus Shaver²¹⁶ (Fig. 52-11) is an inflatable device that removes all accumulated mucus and biofilm within the endotracheal tube. Those prototypes, although currently not commercialized, offer novel insights into future designs and provide research directions to improve removal of secretions.

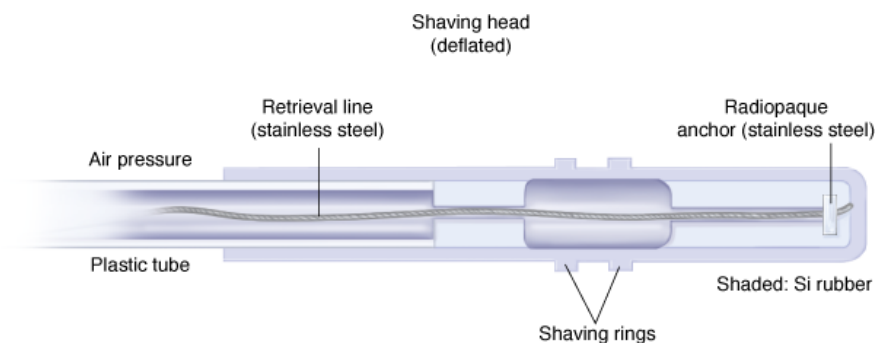
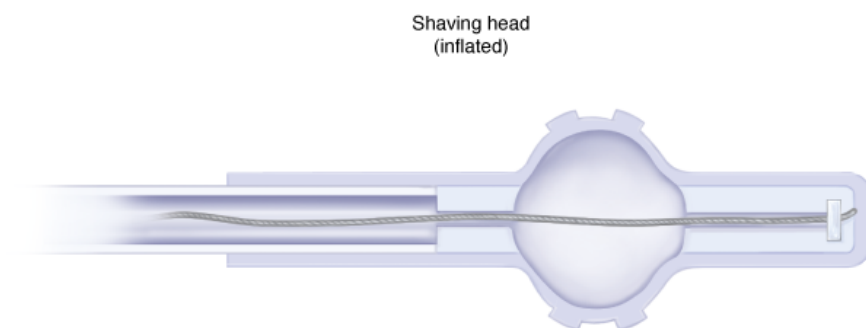
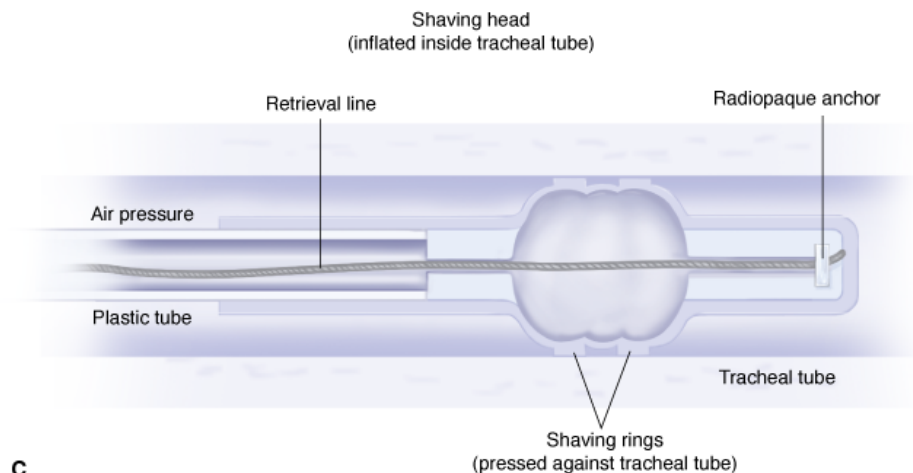
Figure 52-10

**A****B**

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The Mucus Slurper. **A.** The Mucus Slurper is a modified Hi-Lo EVAC tracheal tube (Mallinckrodt Medical, Athlone, Ireland) in which the built-in evacuation lumen is extended beyond the cuff and connected to a vinyl hollow ring with eight holes. Suction at the distal tracheal tube tip is intermittent and automatic. **B.** The cuff of the Mucus Slurper is positioned at the very tip to reduce accumulation of mucus between the cuff and the tip of the Mucus Slurper. When the cuff is inflated, the eight suction holes has no contact with the tracheal mucosa. (Used, with permission, from Kolobow et al²¹⁴ and Li Bassi et al.²¹⁵)

Figure 52-11

**A****B****C**

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The Mucus Shaver. The Mucus Shaver is a 28-cm-long plastic tube with a 2-cm-long silicone inflatable balloon at its tip, with two or more 1.0-mm-wide, 0.5-mm-thick silicone rubber “shaving rings” (A and B). The device is inserted into the artificial airway up to its tip; then the balloon is inflated sufficiently to force the two shaving rings firmly against the internal wall (C). Finally, while inflated, the Mucus Shaver is gently retrieved to remove all accumulated mucus and biofilm. (Used, with permission, from Kolobow et al.²¹⁶)

Another novel device that detects airway secretions through the analysis of airway sounds was recently described.²¹⁷ Although further experimental and clinical evaluation of the device is needed, analysis of airway sounds to detect retained secretions is intriguing and could be a valuable option to signal the need for suctioning.

Summary and Conclusion

The airway lining fluid, composed of mucus and periciliary fluid, overlies the airway epithelium and constitutes a first-line of defense. Mucus is continuously cleared from the airways by cilia and by a two-phase gas–liquid mechanism. Ventilated patients often retain secretions because of mucus hypersecretion and impaired clearance. The effects of gravitational force and ventilator settings on mucus transport are seldom considered, but may be major mechanisms of mucus retention in ventilated patients; these theories require comprehensive evaluation in future studies. Endotracheal suctioning is an essential procedure to remove accumulated mucus within large airways, but can cause discomfort and several complications. Further research is needed to identify the proper indications for suctioning and to improve the technology and safety of the procedure. Manual hyperinflation is based on the principles of mucus transport by means of a two-phase gas–liquid mechanism and, in selected patients, can be useful to centralize retained secretions. Unfortunately, most of the quantitative techniques used to assess mucus clearance are difficult to apply in ventilated patients, and consequently surrogate end points are commonly used to measure the response to interventions, increasing the potential for negative studies and hampering progress in the field.

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